Brazil
Administrative Council for Economic Defense (CADE)

Background

Any initiative that aims at the implementation of computational tools must deal with underlying issues that are usually neglected.

Such issues are core to the challenges faced by the Brazilian competition authority (Administrative Council for Economic Defense, or CADE) as they involve values and rules computational tools can potentially affect. Therefore, we must look for strategies to ensure the computational models capture our values and rules, understand what we did and, above all, understand what our actual objectives with such tools are. This would be the "alignment problem," one of the most relevant questions of our time: on the one hand, there are technologies of pervasive use and opaque constitution, and, on the other, the values and rules essential to a competition authority's institutional actions. Preventing such misalignment means avoiding potentially deleterious results for the society and the economy.

This is because several of these models have a peculiar characteristic: their continued use, without revision, can reinforce and potentiate preexisting problems – such as biases – as the models are subject to feedback loops. In other words, starting from a reality in which there are inequities – which are usually reflected in the databases adopted by these models – these systems produce results that reflect the original biases, and, as the results will return to the system as input for future analyzes and decisions, crystallizing inequities in a vicious cycle that confirms the same biases. Hence, these models not only mimic but, in the long term, can also change the reality they came from as they consolidate and expand the preexisting inequities.

CADE's experience

The Brazilian competition authority has considerable experience in the development of data mining and screening techniques to enhance the detection of signs of bid-rigging (computational tools for improving detection). The authority's goals with them are initiating new investigations and identifying markets that are more susceptible to collusion. As an example, CADE is currently pursuing an ongoing investigation whose developments drew on the use of screening and data mining techniques entirely.[2]

Even though these tools still strive for accurate results, strategies and heuristics have been developed to avoid the risk of false positives and/or false negatives, such as the use of the bag of screening models combined with data mining.

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[2] Based on the screening and data mining results, the authority identified a suspicion of collusion in public tenders, which lead to down raids in 14 companies.

In addition, the authority develops tools aimed at speeding up investigations by automating procedural phases. Based on the analysis of case handlers' workflow vis-à-vis the time spent on each activity, it is possible to see what tools can potentially be automated (computational tools for improving efficiency).

Finally, the use of computational tools for detection activities brought positive externalities to the organization as the tools also collaborated with the work of other areas. For example, CADE's merger units adopted applications developed for research purposes, which in its turn, created new necessities that called for new solutions. That is, the merger units felt the need for additional functionalities to analyze graphs on corporate relationships between companies. Therefore, the tool had to be further improved, bringing about more advances.

For devising these tools, various internal units have engaged in constant dialogue to define common parameters and avoid overlaps between their activities. For instance, (i) the Office of the Superintendent General, through its unit dedicated to the Project Brain, is responsible for conceiving techniques for cartel investigation; (ii) the IT Unit supports the development of initiatives for the automation of internal activities; and (iii) the Department of Economic Studies applies methods to carry out sectoral studies and specialized analyses.

Next steps

CADE is evaluating the development of tools based on artificial intelligence for detecting signs of cartels and improving internal investigation processes.

Besides the mentioned alignment problem, the challenges[3] for detecting signs of cartels are related to the following: AI training data, considering the reduced number of convicted cases; the developed techniques, given the institutional differences of jurisdictions, which can make it difficult to adapt techniques to each local reality; and the transparency of techniques, especially when its results are used for imposing sanctions.

For the second aim, task automation is likely to occur, given the existence of well-known mapped workflows and standardized activities, which would increase significantly the efficiency of the analysis of the work performed, especially the analysis of electronic documents and evidence.

The decision-making process has multiple and complex steps, such as data generation, construction of training bases, the definition of model parameters, tests, and implementation. Hence, it is important to stress that, for the decision-making process, task automation can create biased chains – potentiated at the various links of the AI system — which require extra caution in its implementation. It is important to take into account AI regulatory initiatives, which

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Thus, if implemented thoughtlessly, the use of AI-based computational tools can strengthen the alignment problem, as well as raise concerns about the transparency of the techniques.